

Skills Summary

Illegal Cancellation and Lost Solutions

Use this reference while completing your worksheet or when studying.

Skill 1 — Factor, never cancel

Rule: you may divide both sides by a quantity only if that quantity can never be zero. A trigonometric factor *can* be zero, so dividing by it silently deletes solutions.

Repair: move everything to one side, factor, and set each factor to zero.

$AB = A \implies A(B - 1) = 0$, never $B = 1$ alone.

Example: Solve $2 \cos^2 \theta = \sqrt{3} \cos \theta$ for $0^\circ \leq \theta < 360^\circ$.

Step 1. The same cosine appears on both sides, which is the tempting moment to divide — so collect and factor instead, keeping that cosine as a factor.

Step 2. $\cos \theta (2 \cos \theta - \sqrt{3}) = 0$: $\cos \theta = 0$ or $\cos \theta = \frac{\sqrt{3}}{2}$. $\Rightarrow \theta = 30^\circ, 90^\circ, 270^\circ, 330^\circ$.

Try it: Solve $2 \sin^2 \theta = \sqrt{2} \sin \theta$ for $0^\circ \leq \theta < 360^\circ$.

check: $0^\circ, 45^\circ, 135^\circ, 180^\circ, 225^\circ, 315^\circ$

Skill 2 — Keeping every solution

Three places solutions go missing:

- taking one square root: $u^2 = k$ means $u = \sqrt{k}$ or $u = -\sqrt{k}$;
- a multiple angle: for $k\theta$, multiply both interval endpoints by k before solving;
- declaring a factor impossible: $\cos \theta = 1$ and $\sin \theta = -1$ are attainable.

Example: Solve $2 \cos 2\theta = \sqrt{3}$ for $0^\circ \leq \theta < 360^\circ$.

Step 1. The angle inside is doubled, so the inside variable travels twice as far — widen its interval before solving, or half the answers never appear.

Step 2. $u = 2\theta$ on $[0^\circ, 720^\circ)$: $\cos u = \frac{\sqrt{3}}{2}$ at $u = 30^\circ, 330^\circ, 390^\circ, 690^\circ$; halve each. $\Rightarrow \theta = 15^\circ, 165^\circ, 195^\circ, 345^\circ$.

Try it: Solve $2 \sin 3\theta = \sqrt{3}$ for $0^\circ \leq \theta < 180^\circ$.

check: $0^\circ, 40^\circ, 140^\circ, 160^\circ$

Skill 3 — Testing for extraneous roots

Squaring is not reversible. If $x = y$ then $x^2 = y^2$, but not the other way round: -1 and 1 have the same square. So squaring both sides can *invent* roots.

Repair: squaring is allowed, but every candidate must then be substituted into the *original* equation and kept only if both sides agree.

Example: Solve $\cos \theta + 1 = \sin \theta$ for $0^\circ \leq \theta < 360^\circ$.

Step 1. Two different ratios and no factor in common, so squaring is the way in — but it obliges me to test every candidate afterwards.

Step 2. Squaring gives $2 \cos \theta (\cos \theta + 1) = 0$, so the candidates are 90° , 270° and 180° . Testing: 90° gives $1 = 1$, 180° gives $0 = 0$, but 270° gives $1 = -1$. $\Rightarrow \theta = \mathbf{90^\circ, 180^\circ}$.

Try it: Solve $\sin \theta + \cos \theta = 1$ for $0^\circ \leq \theta < 360^\circ$.

Check: $0 = \theta$:xepq